Generic Fibers and Functional Dimension of Multi-Head Attention

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Abstract

Weight-space symmetries are pervasive across neural architectures, where symmetries often generate positive dimensional families of parameter tuples that realize the same function. Multi-head self-attention has two known symmetries, including linear changes of basis within Q-K and O-V products and permutations of heads. We prove that for generic parameters, multi-head self-attention has no additional hidden symmetries, characterizing the generic fibers of the parametrization. As a consequence, we derive a formula for the functional dimension of multi-head attention. While an architecture's number of parameters is often used as a measure of expressivity as models are scaled, our functional dimension formula demonstrates when this heuristic is valid and when it is less reliable. We study a setting where the number of attention heads is varied while keeping the number of parameters fixed, and experimentally demonstrate that the model's memorization capacity in bits per parameter varies with the number of heads, but that most of this variation is explained away when the bits memorized is normalized by the functional dimension instead of the number of parameters.

1. Introduction

Multi-head self-attention is a fundamental building block of the transformer architecture in modern machine learning, used in large language models (Brown et al., 2020; Chowdhery et al., 2023), and other areas of modern generative AI, including vision and multimodal models (Dosovitskiy, 2020). In self-attention, different choices of the query, key, value, and output matrices can give rise to the same input-output function. The collection of all such parameter choices is called the *fiber* over that function. In many neural network architectures, such fibers often arise from parameter symmetries, namely transformations of the weights that preserve the represented function (Zhao et al., 2026). For attention layers, some symmetries are already apparent from the architecture. For instance, for a single head, the value and output matrices appear through the product $W_O W_V$, while the key and query matrices appear through $W_K^\top W_Q$. Hence, invertible changes of basis in the corresponding intermediate spaces leave the head unchanged. In the multi-head case, there is also the discrete symmetry of permuting heads, since their outputs are summed. This leads to the question studied in this paper. Do these evident symmetries describe the generic fiber, or are there further redundancies in the parametrization?

Answering this question is important for estimating a model's effective capacity. Scaling laws typically use the number of parameters as the model-size axis (Kaplan et al., 2020). Parameter count is also frequently used as a measure of capacity; for instance, studies of language model memorization make this explicit by measuring memorization capacity in bits per parameter (Morris et al., 2025)(Allen-Zhu & Li, 2024). However, when fibers are positive-dimensional, certain directions in weight space do not change the represented function at all, and while these redundancies are included in the number of parameters, they do not contribute to the local dimension of the function space. We therefore argue that functional dimension is a more principled quantity to use when comparing models.

Beyond capacity estimation, the geometry of the generic fiber influences the geometry of the loss landscape. Since the training objective depends on the weights through the represented function, the loss is constant along any fiber. Thus, in architectures with positive-dimensional fibers, we have continuous flat directions in the loss landscape. Furthermore, it is well understood that flat directions in the loss landscape dictate conserved quantities governing gradient flow (Kunin et al., 2021; Zhao et al., 2023). Thus, characterizing all symmetries is a necessary step towards characterizing these conserved quantities and, more generally, understanding the training dynamics of neural networks.

While the fibers are well understood for a single head of

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attention (Henry et al., 2025), to the best of our knowledge, no work has been done characterizing the fibers of multi-head attention.

Organization. In section 2, we characterize the generic fibers of a single layer of multi-head self-attention (with a proof outline in section 3), and use this characterization to derive a functional dimension formula as a corollary. In section 4, we demonstrate experimentally that varying the number of heads at a fixed number of parameters changes memorization capacity, and that normalizing by functional dimension largely explains this variation.

2. Setup and Main Result

Fix an embedding dimension d, head dimensions $d_v, d_a \geq 1$, and number of heads k. We note that typically $d_v = d_a = \frac{d}{k}$, but we work with a more general hyperparameter setup. For each head $t \in [k]$, let $W_O^{(t)} \in \mathbb{R}^{d\times d_v}$, $W_V^{(t)} \in \mathbb{R}^{d_v \times d}$, $W_K^{(t)} \in \mathbb{R}^{d_a\times d}$, and $W_Q^{(t)} \in \mathbb{R}^{d_a \times d}$, and define the per-head products $V_t := W_O^{(t)} W_V^{(t)}$ $A_t := (W_K^{(t)})^\top W_Q^{(t)}$. Given input $X \in \mathbb{R}^{d\times T}$ with T tokens, multi-head softmax attention is

$$\mathrm{MHA}(X) := \sum_{t=1}^{k} V_t X \mathrm{SoftMax}_{\mathrm{col}}(X^\top A_t X), \qquad (1)$$

where $\mathrm{SoftMax}_{\mathrm{col}}$ applies softmax to each column of its argument.

The parameter space is thus defined as

$$\mathcal{P} := (\mathbb{R}^{d\times d_v} \times \mathbb{R}^{d_v\times d} \times \mathbb{R}^{d_a\times d} \times \mathbb{R}^{d_a\times d})^k. \qquad (2)$$

We write MHA_p for the function defined by $p \in \mathcal{P}$ and study the fibers of $p \mapsto \mathrm{MHA}_p$. We focus on the generic setting, excluding certain degenerate parameter choices which form an *exceptional set* $\mathcal{N} \subseteq \mathcal{P}$. Importantly, we construct $\mathcal{N}$ so that the probability of sampling a parameter from this set is 0. More precisely, we choose $\mathcal{N}$ to be the following Zariski-closed set:

$$\mathcal{N} := \bigcup_{t=1}^{k}\{p : \mathrm{rank}(V_t) < d_v\} \cup \bigcup_{t=1}^{k}\{p : \mathrm{rank}(A_t) < d_a\} \cup \bigcup_{t<t'}\{p : A_t = A_{t'}\}.$$

We call p' generic if $p' \notin \mathcal{N}$. In the following theorem, we describe the generic fibers of MHA.

Theorem 2.1. *Assume that there are at least 3 tokens in the sequence ($T \geq 3$), the embedding dimension is at least 2 ($d \geq 2$), and the attention and value embedding dimensions are at most as large as the embedding dimension ($1 \leq d_v, d_a \leq d$).*

For $p' \in \mathcal{P} \setminus \mathcal{N}$ and $p \in \mathcal{P}$, we have $\mathrm{MHA}_p = \mathrm{MHA}_{p'}$ if and only if there exists a permutation $\rho \in S_k$ and invertible matrices $G_t \in \mathrm{GL}_{d_v}(\mathbb{R})$, $H_t \in \mathrm{GL}_{d_a}(\mathbb{R})$ for $t \in [k]$ such that

$$W_O^{(t)} = W_O'^{(\rho(t))} G_t, \qquad W_V^{(t)} = G_t^{-1} W_V'^{(\rho(t))}, \qquad (3)$$

$$W_K^{(t)} = H_t W_K'^{(\rho(t))}, \qquad W_Q^{(t)} = H_t^{-\top} W_Q'^{(\rho(t))}. \qquad (4)$$

Proof. An outline of the proof is in Section 3, and the complete proof can be found in Appendix A. ☐

As noted in the introduction, characterizing the fibers leads to a formula for functional dimension. We define functional dimension rigorously below.

Definition 2.2 (Functional dimension)**.** Let $\Phi : \mathcal{P} \to \mathcal{F}$ be the parametrization map sending a parameter to the function it represents. The *local functional dimension* of Φ at a regular parameter p is

$$\mathrm{fdim}_p(\Phi) = \mathrm{rank}\, D\Phi_p,$$

where $D\Phi_p$ denotes the differential of Φ at p.

Equivalently, $\mathrm{fdim}_p(\Phi)$ is the dimension of the tangent space, when the tangent space is defined. If this quantity is constant on a generic subset of $\mathcal{P}$, we call the common value the *generic functional dimension*.

The fiber dimension theorem (Lee, 2013) states that the generic functional dimension is equal to the number of parameters minus the generic fiber dimension. In this sense, the fiber dimension can be interpreted as the number of *redundant parameters* and the functional dimension as the number of *effective parameters*. Moving forward, we use the notation P_{eff} for the functional dimension to emphasize that it is a notion of effective parameter count, and use the fiber dimension theorem in the following Corollary to derive a formula for it.

Corollary 2.3. *The generic fiber of the map $p \mapsto \mathrm{MHA}_p$ has dimension $k(d_v^2 + d_a^2)$. Therefore, by the fiber dimension theorem,*

$$P_{\text{eff}} = \dim \mathcal{P} - k(d_v^2 + d_a^2) = k\big(2d(d_v + d_a) - d_v^2 - d_a^2\big).$$

When $d_v = d_a = d/k$, this simplifies to $P_{\text{eff}} = 4d^2 - \frac{2d^2}{k}$.

We hypothesize that, for an attention-only transformer with skip connections, the per-layer symmetries remain independent across layers and cross-layer interactions do not enlarge the fibers. We formalize this in the conjecture below.

Conjecture 2.4. For an L-layer multi-head attention-only transformer with skip connections, and no Multi-layer Perceptrons or additional normalizations, the generic fiber dimension is L times the single-layer fiber dimension. Equivalently, the functional dimension is

$$P_{\text{eff}} = L\left(4d^2 - \frac{2d^2}{k}\right). \quad (5)$$

3. Proof Overview

We provide a proof outline of Theorem 2.1 in this section, while the full argument can be found in Appendix A. The proof has two parts. First, we show that equality of two multi-head attention functions generically forces equality of the per-head product parameters $V_t = W_O^{(t)} W_V^{(t)}$ and $A_t = (W_K^{(t)})^\top W_Q^{(t)}$ up to a permutation of heads. Second, we lift equality of these products back to equality of the original factors up to the expected invertible basis changes.

We perform this first step by reducing the SoftMax to sigmoid functions. This is done by evaluating the attention layer on input sequences consisting of two repeated tokens. Along this family of inputs, each head contributes a term of the form

$$V_t w \sigma(w^\top A_t u \cdot \tau + b),$$

where the slope $w^\top A_t u$ depends on the attention product A_t. For generic u, w, distinct attention products produce distinct nonzero slopes. Inspired by (Fefferman, 1994), we perform analytic continuation and separate the heads based on their poles, which depend on these slopes. This then implies that terms cannot cancel unless the two functions $\mathrm{MHA}_{p'}$ and MHA_p have matching poles. From this, we conclude that they must have heads with matching slopes up to permutations, and deduce that we can identify the A_t's and V_t's up to permutations of heads.

We then, prove that, modulo head permutations, the fibers consist of exactly the usual matrix-factorization symmetries. If $W_O W_V = W'_O W'_V$ has rank d_v, then the two factorizations differ by a unique element of GL_{d_v}. The same argument applied to $W_K^\top W_Q$ gives the GL_{d_a} symmetry for the key-query factors. Therefore, the generic fibers consist of k independent value-output symmetries, k independent key–query symmetries, and a finite head permutation. Since finite permutations do not affect the dimension, the generic fiber dimension is $k(d_v^2 + d_a^2)$.

4. Experiments

Experimental question. We design an experiment to test the implications of the functional dimension formula on memorization scaling-laws. Holding total number of parameters fixed, we vary only the number of attention heads. If parameter count alone captured model capacity, varying the number of heads should leave the bits memorized per parameter unchanged, as suggested by (Morris et al., 2025). Our theory instead predicts that the bits memorized per parameter will increase as the number of heads increases, as a result of the functional dimension increasing.

Setup. We train 2-layer attention-only transformers with no MLP, no layer normalization, and no biases to mimic the proof's setting of Corollary 2.3 and Conjecture 2.4. We use $d = 32$, vocabulary size $V = 128$, and sequence length $S = 16$. Furthermore, we consider the standard case where $d_a = d_v = d/k$. The total parameter count is then

$$n_{\text{params}} = Vd + Sd + L \cdot 4d^2 = 12{,}800,$$

which is independent of the number of heads k by construction. Since the model has $L = 2$ layers, we invoke Conjecture 2.4 to obtain the effective parameter count $P_{\text{eff}}(k) = n_{\text{params}} - 2Ld^2/k$, ranging from $8{,}704$ (k=1) to $12{,}544$ (k=16).

We train on $n_{\text{train}} = 16{,}384$ uniformly random token sequences. Following (Morris et al., 2025), we measure memorization capacity in bits by computing the cross-entropy loss on the training set and convert the per-token loss reduction (relative to the uniform baseline $\log_2 V = 7$ bits) to total bits memorized. We sweep $k \in \{1, 2, 4, 8, 16\}$ with 3 random seeds each. Learning rates and initialization scales are tuned carefully for each head count (see supplementary material in Appendix B for details) and training uses Adam with BF16 mixed precision.

Results. Table 1 and Figure 1 summarize the findings. When normalizing by the number of parameters, as our theory predicted, the bits memorized per parameter increases with the number of heads, indicating that the number of parameters alone is not an ideal measure of capacity. Furthermore, after subtracting the generic fiber dimension (i.e. the "redundant parameter" count) and normalizing by effective parameter count instead, the variation is much smaller, further following our predictions. Bits per parameter ranges from 1.83 to 2.83, a 55% increase, whereas bits per effective parameter ranges from 2.69 to 2.89, only 7% variation. Thus, the dependence of bits per parameter on the number of heads is largely explained by the changing functional dimension. This discrepancy is much larger in architectures with a smaller number of heads, while for a larger number of heads, we find that the number of parameters is a more reasonable estimate of the functional dimension.

We believe that the remaining slight monotone trend in the bits per effective parameter is likely due to differing optimization dynamics arising from a different number of heads. However, this discrepancy may also be a result of imperfect hyperparameter tuning. Nevertheless, the difference in results between the axes of bits per parameter and bits per effective parameter supports the claim that, in this setting,

Table 1. Memorization capacity across head counts. All configurations have $n_{\mathrm{params}} = 12{,}800$. Mean $\pm$ std over 3 seeds.

k	P_{eff}	Bits	bpp	bpep
1	8,704	23,372	1.83	2.69
2	10,752	30,235	2.36	2.81
4	11,776	33,420	2.61	2.84
8	12,288	35,316	2.76	2.87
16	12,544	36,224	2.83	2.89

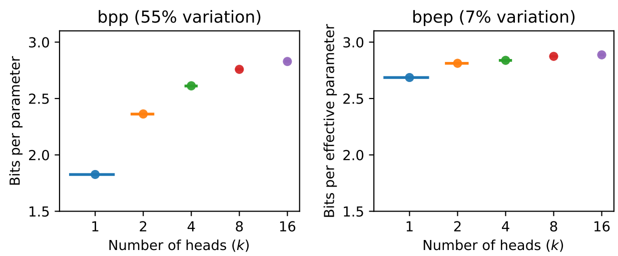

Figure 1. Final bits per parameter (left) and bits per effective parameter (right) versus number of heads. Each dot is one seed; horizontal bars show means. Normalizing by the theory-derived functional dimension greatly reduces the spread.

functional dimension is a more architecture-aware measure of capacity than the number of parameters.

5. Related Work

Transformers and attention geometry. The closest predecessor to the present work is (Henry et al., 2025), which studies the geometry of lightning self-attention, also called linear self-attention. Their work explicitly leaves multi-head variants among the architectural extensions still to be addressed. Subsequent work by (Alexandr et al., 2026) develops a complementary algebraic-invariant perspective on lightning self-attention, studying polynomial coefficient constraints such as low-rank, Chow-type, Veronese-type, and resultant-based relations. The broader program of using algebraic geometry to study neural-network function spaces is surveyed and advocated by (Marchetti et al., 2025).

Symmetries and identifiability. Parameter symmetries are well understood to lead to a lack of identifiability in neural networks. For classical sigmoidal networks, (Fefferman, 1994) use complex analysis to identify all of the architecture's weight-space symmetries, characterizing the fibers. (Vlačić & Bölcskei, 2022) extend this line of work to a broader family of sigmoidal nonlinearities. In modern deep learning, parameter-space symmetries have been connected to loss-landscape geometry, optimization, model merging, and learning dynamics, as explored in the survey of (Zhao et al., 2026). Examples of this include symmetry-induced conservation laws in learning dynamics (Kunin et al., 2021) and model merging modulo permutation symmetries (Ainsworth et al., 2023).

Functional dimension and hidden symmetries. Functional dimension asks how many local degrees of freedom a parameterized architecture actually realizes in function space after quotienting out redundant directions. (Grigsby et al., 2023) show that ReLU networks can also admit hidden symmetries and empirically estimate functional dimension across architectures. (Grigsby et al., 2025) develop the functional-dimension theory for feedforward ReLU networks in detail, emphasizing that functional dimension can be inhomogeneous across parameter space for neural networks, contrasting with our findings that functional dimension is constant across a generic subset of parameter space.

Scaling laws and capacity axes. Neural scaling laws commonly express performance as a function of model size, dataset size, and compute. In the language-model setting, (Kaplan et al., 2020) use the number of non-embedding parameters as the model-size coordinate and find that loss scales predictably with this quantity, while depending weakly on architectural choices. Our work is complementary as we study this claim more carefully and explore a setting where parameter count is fixed but the local dimension of the represented function class changes. The resulting functional dimension gives a symmetry- corrected model-size coordinate for multi-head attention, which can potentially be used as an improved axis for scaling laws. Our memorization experiments follow (Morris et al., 2025), who measure capacity in bits per parameter on synthetic data with the possibility for generalization eliminated. This is closely related to (Allen-Zhu & Li, 2024), who study structured factual storage and report an approximately 2-bit-per-parameter capacity law. Together, these works provide baselines for compression-style memorization and factual knowledge storage. We use a similar random-sequence setup to (Morris et al., 2025), but normalize by the functional dimension predicted by our theory rather than only by the number of parameters, adding more nuance to their claim that memorization capacity should be thought of as linear in the number of parameters.

Impact Statement

This paper presents theoretical and empirical work on understanding the structure of neural network parameter spaces. As our work is theoretical, we do not believe that there are societal impacts which must be specifically highlighted here.

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A. Proof of Theorem 2.1

The argument first reduces softmax attention to a family of sigmoids by evaluating on inputs with two repeated tokens, then applies analytic separation ideas inspired by (Fefferman, 1994) to identify the per-head products up to head permutation, and concludes by studying the fibers of the per-head products to identify the fibers with respect to the parameter space.

For a single head, softmax attention can be expressed as follows

$$\mathrm{Attn}(V, A)(X) := VX\mathrm{SoftMax}_{\mathrm{col}}(X^\top AX). \tag{6}$$

The following lemma shows how Equation 6 can be reduced to an equation involving sigmoids instead of a softmax.

Lemma A.1. *Let $u, v \in \mathbb{R}^d$ and let $r, s \geq 1$ satisfy $r + s = T$. For $\tau > 0$, let $X_{u,v}(\tau) \in \mathbb{R}^{d\times T}$ have its first r columns equal to $\sqrt{\tau}u$ and last s columns equal to $\sqrt{\tau}v$, and let $b := \log(r/s)$. Then for any query column $x_j = \sqrt{\tau}u$,*

$$\frac{1}{\sqrt{\tau}}\mathrm{Attn}(V, A)(X_{u,v}(\tau))_{:,j} = Vv + V(u - v)\,\sigma\big((u - v)^\top Au\tau + b\big)$$

Proof. Fix a query column $x_j = \sqrt{\tau}u$, and key rows $x_a = \sqrt{\tau}u^\top$ and $x_b = \sqrt{\tau}v^\top$. Substituting $X_{u,v}(\tau)$ for X in $\mathrm{SoftMax}_{\mathrm{col}}(X^\top AX)_{i,j}$ simplifies as follows

$$\mathrm{SoftMax}_{\mathrm{col}}(X_{u,v}(\tau)^\top AX_{u,v}(\tau))_{a,j} = \frac{e^{\tau u^\top Au}}{re^{\tau u^\top Au} + se^{\tau v^\top Au}} = \frac{1}{r}\sigma\left(\left((u - v)^\top Au\right)\tau + b\right) \tag{7}$$

Using the property that columns of $\mathrm{SoftMax}_{\mathrm{col}}$ sum to 1, we have

$$s\mathrm{SoftMax}_{\mathrm{col}}(X_{u,v}(\tau)^\top AX_{u,v}(\tau))_{b,j} + r\mathrm{SoftMax}_{\mathrm{col}}(X_{u,v}(\tau)^\top AX_{u,v}(\tau))_{b,j} = 1. \tag{8}$$

Substituting equation (8) into (6), we obtain

$$\frac{1}{\sqrt{\tau}}\mathrm{Attn}(V, A)(X_{u,v}(\tau))_{:,j} = rVu\mathrm{SoftMax}_{\mathrm{col}}(X^\top AX)_{a,j} + Vv\left(1 - r\mathrm{SoftMax}_{\mathrm{col}}(X^\top AX)_{b,j}\right). \tag{9}$$

As required, this simplifies to

$$\frac{1}{\sqrt{\tau}}\mathrm{Attn}(V, A)(X_{u,v}(\tau))_{:,j} = Vv + V(u - v)\sigma\left(\left((u - v)^\top Au\right)\tau + b\right).$$

$\square$

The following lemma proves a linear-independence-like property of shifted sigmoid functions with different amounts of horizontal scaling.

Lemma A.2. *Fix $b \in \mathbb{R} \setminus \{0\}$. Let $\lambda_1, \ldots, \lambda_m \in \mathbb{R} \setminus \{0\}$ be pairwise distinct. Let $I \subseteq \mathbb{R}$ be a nonempty open interval. If*

$$c + \sum_{j=1}^{m} \alpha_j\sigma(\lambda_j t + b) = 0 \qquad \textit{for all } t \in I$$

for some $c, \alpha_1, \ldots, \alpha_m \in \mathbb{R}$, then $c = \alpha_1 = \cdots = \alpha_m = 0$. The same conclusion holds coordinatewise if $c, \alpha_1, \ldots, \alpha_m$ are vectors in $\mathbb{R}^q$.

Proof. It suffices to prove the scalar case, since the vector-valued case follows by applying the scalar result to each coordinate. Consider the meromorphic function

$$F(z) = c + \sum_{j=1}^{m} \alpha_j\sigma(\lambda_j z + b) \tag{10}$$

on $\mathbb{C}$. The function is holomorphic on the complement of its poles and suppose that it vanishes on the real interval I. By the identity theorem, F vanishes identically as a meromorphic function.

The poles of $\sigma(\lambda_j z+b)$ occur when $1+\exp(-\lambda_j z-b)=0$, equivalently at $z=((2n+1)\pi i-b)/\lambda_j$ for $n\in\mathbb{Z}$. We claim that these pole sets are disjoint for distinct j. Suppose by way of contradiction that two poles coincide for parameters λ_j and λ_ℓ. Then, there exist integers n and m where $\big(\lambda_\ell(2n+1)-\lambda_j(2m+1)\big)\pi i=(\lambda_\ell-\lambda_j)b$. The left-hand side is purely imaginary and the right-hand side is real. Hence both sides must be zero. Since $b\neq 0$, this gives $\lambda_\ell=\lambda_j$, contradicting pairwise distinctness unless $j=\ell$.

Thus each $\sigma(\lambda_j z+b)$ has poles that occur in no other term of F. At such a pole, the principal part of F is nonzero unless $\alpha_j=0$. Since F is identically zero as a meromorphic function, it has no nonzero principal parts. Therefore $\alpha_j=0$ for all j, and then $c=0$. □

We now prove in the following proposition that two equal multi-head attention functions must have the same parameter products V_t and A_t up to a permutation of heads t.

Proposition A.3. *Let $p,p'\in\mathcal{P}$ satisfy $\mathrm{MHA}_p=\mathrm{MHA}_{p'}$. Write $V_t=W_O^{(t)}W_V^{(t)}$, and $A_t=(W_K^{(t)})^\top W_Q^{(t)}$ for the products of p, and write $\widetilde{V}_j=W_O'^{(j)}W_V'^{(j)}$, $\widetilde{A}_j=(W_K'^{(j)})^\top W_Q'^{(j)}$ for the products of p'.*

Assume that $\mathrm{rank}(\widetilde{V}_j)=d_v$, $\mathrm{rank}(\widetilde{A}_j)=d_a$ for all j, and that the matrices $\widetilde{A}_1,\dots,\widetilde{A}_k$ are pairwise distinct. Then there exists a unique permutation $\rho\in S_k$ such that $A_t=\widetilde{A}_{\rho(t)}$ and $V_t=\widetilde{V}_{\rho(t)}$ for all $t\in[k]$.

Proof. Since $T\geq 3$, choose $r=T-1$ and $s=1$. Then $b:=\log(r/s)=\log(T-1)\neq 0$. For $u,v\in\mathbb{R}^d$, set $w:=u-v$. Apply Lemma A.1 to inputs containing r copies of $\sqrt{\tau}u$ and one copy of $\sqrt{\tau}v$. For a query token of type u, equality $\mathrm{MHA}_p=\mathrm{MHA}_{p'}$ gives, for all $\tau>0$,

$$0=\sum_{t=1}^{k}\left[V_t v+V_t w\sigma\big(w^\top A_t u\cdot\tau+b\big)\right]-\sum_{j=1}^{k}\left[\widetilde{V}_j v+\widetilde{V}_j w\sigma\Big(w^\top\widetilde{A}_j u\cdot\tau+b\Big)\right]. \tag{11}$$

Let $\mathcal{B}$ be the finite set of distinct nonzero matrices appearing among the A_t and the $\widetilde{A}_j$:

$$\mathcal{B}:=\{B\neq 0: B=A_t \text{ for some } t\}\cup\{B\neq 0: B=\widetilde{A}_j \text{ for some } j\}.$$

For $B\in\mathcal{B}$, define

$$\Gamma_B:=\sum_{\{t:A_t=B\}}V_t-\sum_{\{j:\widetilde{A}_j=B\}}\widetilde{V}_j. \tag{12}$$

Then (11) can be rewritten as the following function of τ

$$0=c(u,w)+\sum_{B\in\mathcal{B}}\Gamma_B w\sigma\big(w^\top Bu\cdot\tau+b\big), \tag{13}$$

where $c(u,w)$ does not vary with τ and is defined as

$$c(u,w):=\left(\sum_{t=1}^{k}V_t-\sum_{j=1}^{k}\widetilde{V}_j\right)(u-w)+\sigma(b)\left(\sum_{\{t:A_t=0\}}V_t\right)w. \tag{14}$$

We now define a space of (u,w) where the slopes in (14) are nonzero and pairwise distinct, and show that this space is nonempty and Zariski-open. Define

$$\Omega:=\left\{(u,w)\in\mathbb{R}^d\times\mathbb{R}^d:\begin{array}{l} w^\top Bu\neq 0 \text{ for all } B\in\mathcal{B},\\ w^\top(B-C)u\neq 0 \text{ for all distinct } B,C\in\mathcal{B}\end{array}\right\}.$$

Indeed, if $B\neq 0$, then the bilinear polynomial $(u,w)\mapsto w^\top Bu$ is not identically zero. Similarly, if $B\neq C$, then $(u,w)\mapsto w^\top(B-C)u$ is not identically zero. The complement of Ω is therefore the zero set of a finite product of nonzero polynomials, so Ω is nonempty and Zariski-open.

For every $(u, w) \in \Omega$, the slopes $\lambda_B(u, w) := w^\top Bu$ for $B \in \mathcal{B}$ are nonzero and pairwise distinct. Applying Lemma A.2 coordinatewise to (14) on the interval $(0, \infty)$ gives $\Gamma_B w = 0$ for every $B \in \mathcal{B}$. Now fix $B \in \mathcal{B}$ and a coordinate ℓ. The function $(u, w) \mapsto e_\ell^\top \Gamma_B w$ is a polynomial that vanishes on Ω. Since Ω is nonempty Zariski-open, it is Zariski dense in $\mathbb{R}^d \times \mathbb{R}^d$, so the polynomial vanishes identically. Therefore, $\Gamma_B w = 0$ for all $w \in \mathbb{R}^d$, and consequently $\Gamma_B = 0$ or every $B \in \mathcal{B}$.

We now show that every attention product $\widetilde{A}_j$ appears among the A_t. Since $\mathrm{rank}(\widetilde{A}_j) = d_a \geq 1$, each $\widetilde{A}_j$ is nonzero, so $\widetilde{A}_j \in \mathcal{B}$. Suppose, for contradiction, that for some j there is no t with $A_t = \widetilde{A}_j$. Since the $\widetilde{A}_j$ are pairwise distinct, the definition of $\Gamma_{\widetilde{A}_j}$ gives $\Gamma_{\widetilde{A}_j} = -\widetilde{V}_j$. But the fact that $\Gamma_B = 0$ gives $\Gamma_{\widetilde{A}_j} = 0$, hence $\widetilde{V}_j = 0$, contradicting $\mathrm{rank}(\widetilde{V}_j) = d_v \geq 1$. Therefore every $\widetilde{A}_j$ appears among the A_t. By the genericity condition, each $\widetilde{A}_j$ is a distinct matrix, it follows that each A_j is also distinct, and furthermore that there exists a unique permutation $\rho \in S_k$ such that $A_t = \widetilde{A}_{\rho(t)}$ for all $t \in [k]$.

Finally, for each t, the matrix $A_t = \widetilde{A}_{\rho(t)}$ occurs once for each multi-head attention function. Therefore $\Gamma_{A_t} = V_t - \widetilde{V}_{\rho(t)}$. Noting again that $\Gamma_B = 0$ for $B \in \mathcal{B}$, gives $V_t = \widetilde{V}_{\rho(t)}$. This proves the claim. $\square$

Lemma A.4. *Let $A, A' \in \mathbb{R}^{d\times r}$, $B, B' \in \mathbb{R}^{r\times d}$ for $r \leq d$, and suppose that $AB = A'B'$ and $\mathrm{rank}(AB) = r$. Then there exists a unique invertible matrix $G \in \mathrm{GL}_r(\mathbb{R})$ such that $A' = AG$ and $B' = G^{-1}B$.*

Proof. The proof is given in (Henry et al., 2025), Lemma 3.1. $\square$

We now combine the above lemmas and propositions to prove our main theorem.

A.1. Proof of Theorem 2.1

Proof. We first verify that $\mathcal{N}$ is a finite union of proper algebraic sets. The conditions $\mathrm{rank}(V_t) < d_v$ and $\mathrm{rank}(A_t) < d_a$ are given by the vanishing of all $d_v \times d_v$ minors of V_t and all $d_a \times d_a$ minors of A_t, respectively. Since the entries of $V_t = W_O^{(t)} W_V^{(t)}$ and $A_t = (W_K^{(t)})^\top W_Q^{(t)}$ are polynomial functions of the parameters, these are algebraic conditions. The conditions $A_t = A_{t'}$ are also polynomial conditions. Each of these algebraic sets is also a proper subset as we can easily choose W_O and W_V so that $W_O W_V$ is rank d_v, choose W_Q and W_K so that $W_K^\top W_Q$ has rank d_a, and also choose distinct rank-d_a matrices for A_t and $A_{t'}$. Hence none of the defining algebraic conditions is identically satisfied on $\mathcal{P}$. Therefore $\mathcal{N}$ is a finite union of proper algebraic sets, and so has Lebesgue measure zero.

Now let $p' \in \mathcal{P} \setminus \mathcal{N}$ and let $p \in \mathcal{P}$ satisfy $\mathrm{MHA}_p = \mathrm{MHA}_{p'}$. Since $p' \notin \mathcal{N}$, the conditions of Proposition A.3 apply. It gives a unique permutation $\rho \in S_k$ such that $A_t = \widetilde{A}_{\rho(t)}$ and $V_t = \widetilde{V}_{\rho(t)}$ for every $t \in [k]$.

We now lift these product equalities to factor equalities. For the value–output factors, apply Lemma A.4. Since $W_O^{(t)} W_V^{(t)} = W_O'^{(\rho(t))} W_V'^{(\rho(t))}$ and the common product has rank d_v, there exists a unique $G_t^{(v)} \in \mathrm{GL}_{d_v}(\mathbb{R})$ such that $W_O^{(t)} = W_O'^{(\rho(t))} G_t^{(v)}$ and $W_V^{(t)} = (G_t^{(v)})^{-1} W_V'^{(\rho(t))}$. For the key–query factors, we also apply Lemma A.4 and similarly find that there exists a unique $G_t^{(a)} \in \mathrm{GL}_{d_a}(\mathbb{R})$ such that $(W_K^{(t)})^\top = (W_K'^{(\rho(t))})^\top G_t^{(a)}$ and $W_Q^{(t)} = (G_t^{(a)})^{-1} W_Q'^{(\rho(t))}$. This proves the forward implication. It also shows $p \notin \mathcal{N}$, because the products of p are obtained from the products of p' by permutation $V_t = \widetilde{V}_{\rho(t)}$ and $A_t = \widetilde{A}_{\rho(t)}$.

Conversely, suppose there exist $\rho \in S_k$ and invertible matrices $G_t^{(v)} \in \mathrm{GL}_{d_v}(\mathbb{R})$ and $G_t^{(a)} \in \mathrm{GL}_{d_a}(\mathbb{R})$ such that $W_O^{(t)} = W_O'^{(\rho(t))} G_t^{(v)}$, $W_V^{(t)} = (G_t^{(v)})^{-1} W_V'^{(\rho(t))}$, and $W_K^{(t)} = G_t^{(a)} W_K'^{(\rho(t))}$, $W_Q^{(t)} = (G_t^{(a)})^{-\top} W_Q'^{(\rho(t))}$.

Then $W_O^{(t)} W_V^{(t)} = W_O'^{(\rho(t))} W_V'^{(\rho(t))}$ and $(W_K^{(t)})^\top W_Q^{(t)} = (W_K'^{(\rho(t))})^\top W_Q'^{(\rho(t))}$. Thus the tth head of p has the same product parameters as the $\rho(t)$th head of p'. Since multi-head attention sums over heads, permuting heads does not change the represented function. Therefore $\mathrm{MHA}_p = \mathrm{MHA}_{p'}$. This proves the theorem. $\square$

B. Experimental Details

Architecture. The model is an attention-only transformer with $L = 2$ layers, $d_{\text{model}} = 32$, learned position embeddings, tied input/output embeddings, a scaled output projection, and no biases, MLP blocks, or layer normalization. This architecture is chosen to match the setup of Theorem 2.1 as closely as possible.

Data. Training sequences consist of $n_{\text{train}} = 16{,}384$ sequences of length $S = 16$, each token drawn uniformly from a vocabulary of size $V = 128$. The total information content is approximately $16{,}384 \times 16 \times 7 \approx 1.84 \times 10^6$ bits, roughly $24\times$ the model's capacity, ensuring that we are in the saturation regime.

Hyperparameter tuning. For each head count, we independently tuned learning rate and initialization standard deviation over a large grid (learning rates in $[10^{-4}, 2 \times 10^{-2}]$, init_std in $[0.05, 0.5]$). The final hyperparameters were k=1: lr = 10^{-3}, init_std = 0.25; k=2: lr = 10^{-3}, init_std = 0.20; k=4: lr = 10^{-3}, init_std = 0.10; k=8: lr = 10^{-3}, init_std = 0.10; k=16: lr = 2×10^{-3}, init_std = 0.35. All runs used 200K steps, batch size 4096, Adam with constant learning rate after a 100-step warmup, and BF16 mixed precision.